



Dry Bean
Insights

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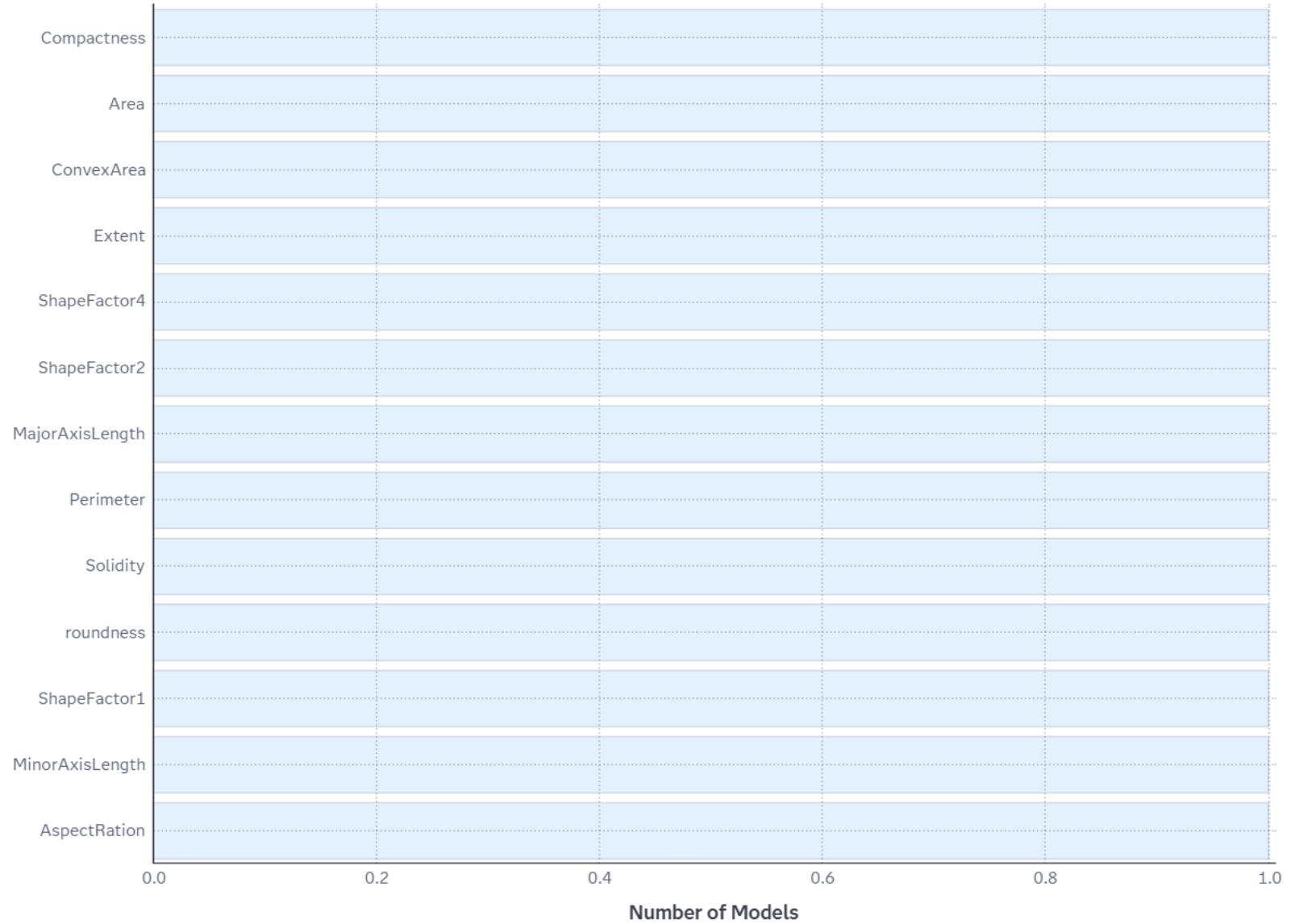
Project Summary

The champion model for this project is Gradient Boosting from the "Pipeline 1" pipeline. The model was chosen based on the KS (Youden) for the Test partition (0.89). The five most important factors are Compactness, Area, AspectRatio, ShapeFactor1, and MinorAxisLength.

Project Target:	Class	Project Champion:	Gradient Boosting
Event Percentage:	19.3667%	Created By:	ai230076@student.uthm.edu.my
Pipelines:	1	Modified:	May 30, 2026, 03:51:13 AM

Project Notes

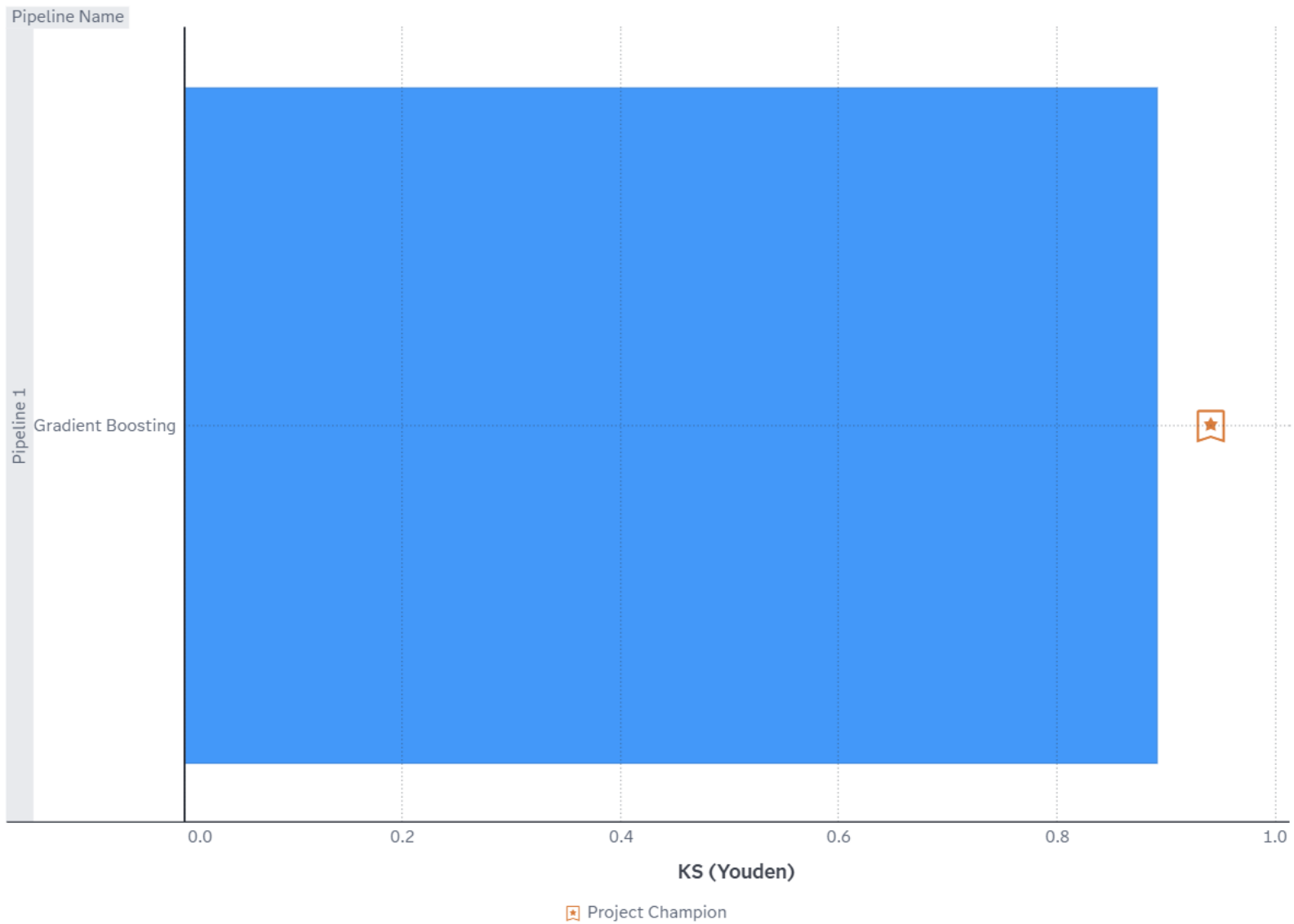
Most Common Variables Selected Across All Models



This plot shows the number of times that an input was deemed an important variable for any model that was used in

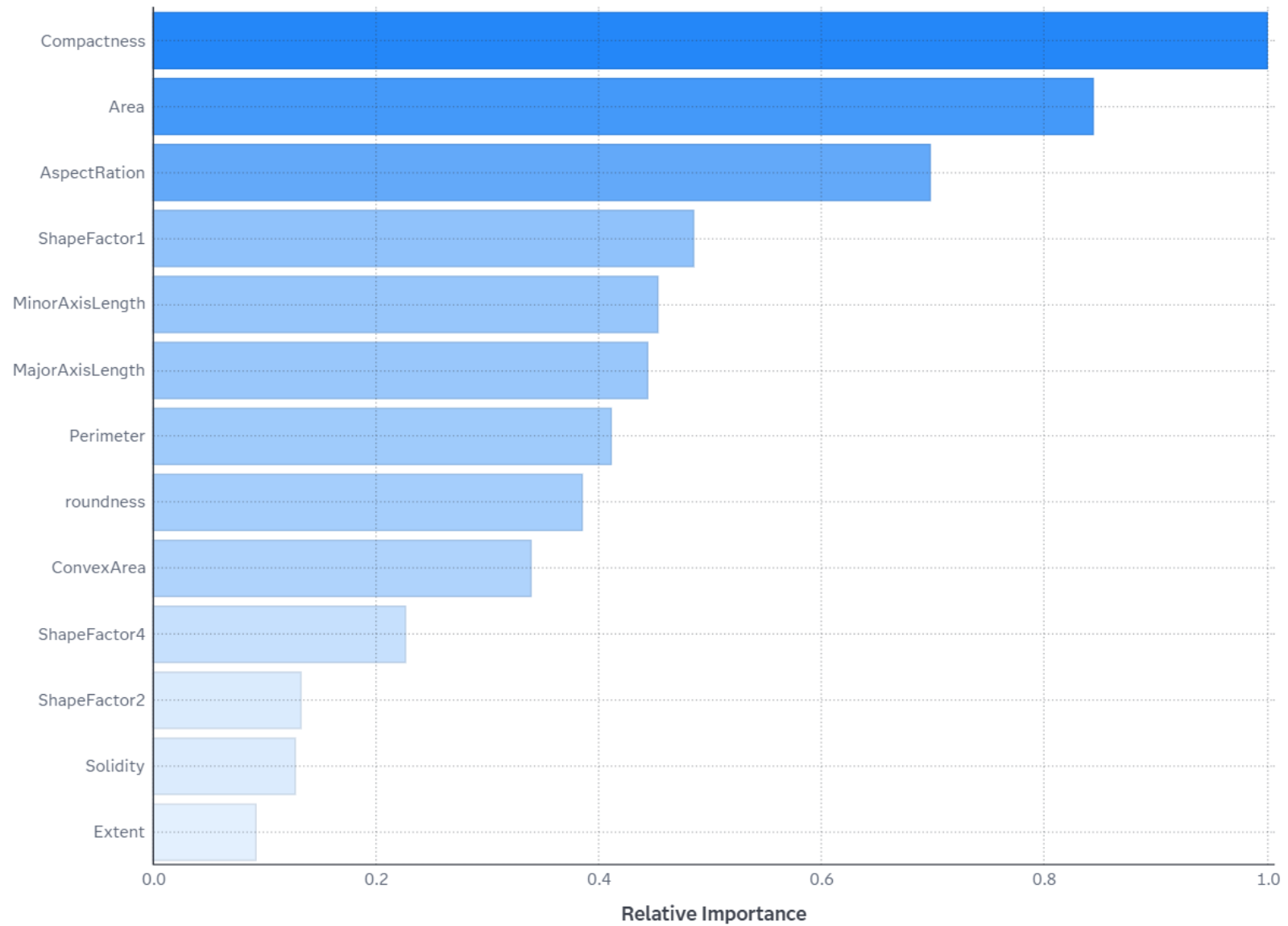
Pipeline Comparison, including the pipeline champions and challenger models. Variable importance is calculated using a surrogate model, a one-level decision tree for each input where the target is the predicted class or value. Inputs with a positive importance value are determined to be important. The most important inputs across the champion and challenger models for this project appear at the top of the plot.

Assessment for All Models



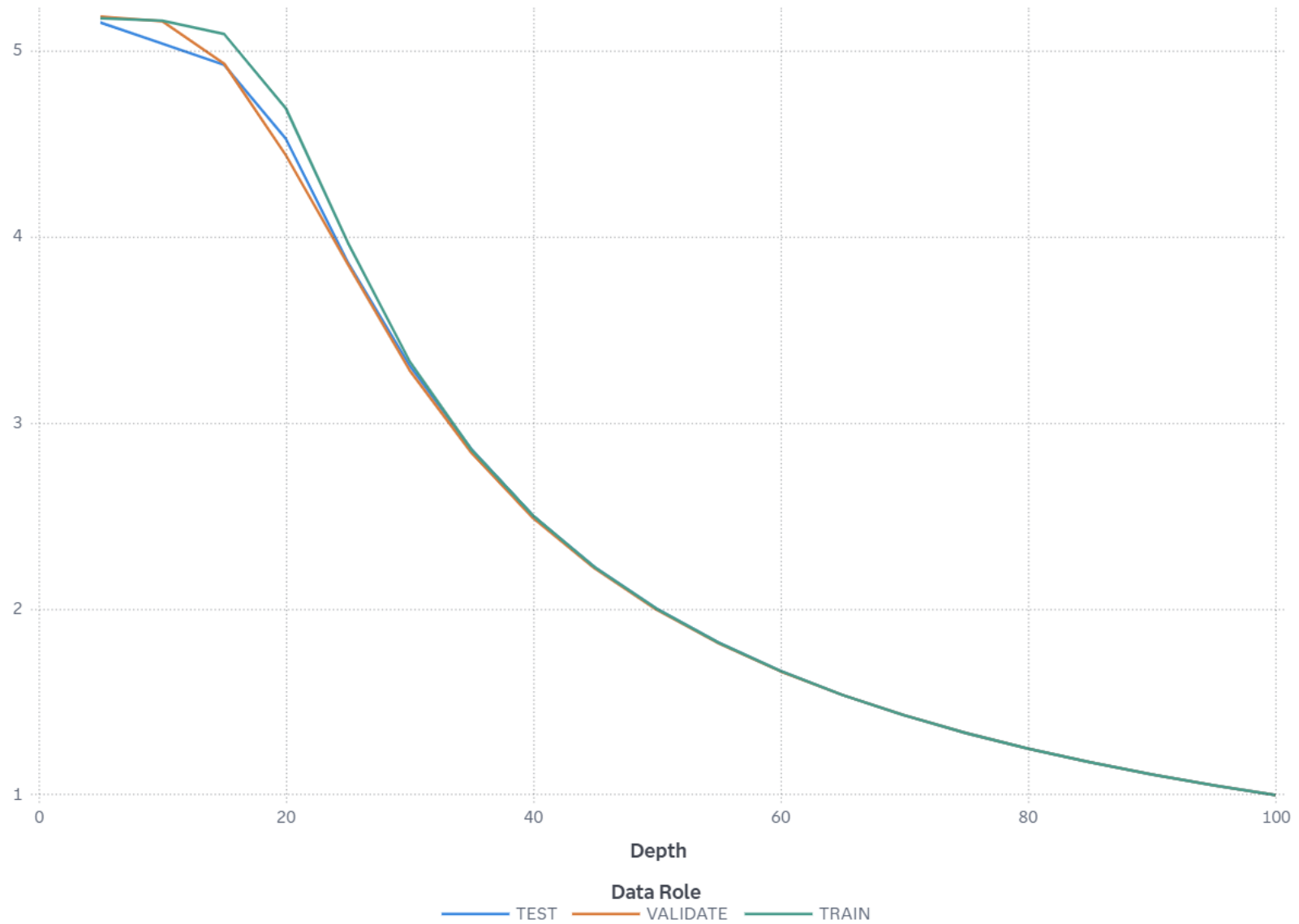
This plot shows the champion from the one pipeline. The overall project champion is the Gradient Boosting model from pipeline "Pipeline 1".

Most Important Variables for Champion Model



This plot shows the 13 most important variables, as determined by the relative importance calculated using the actual model. The most important input for this model is Compactness. The input Area has a relative importance of 0.84, for example, which means it is 0.84 times as important as Compactness.

Cumulative Lift for Champion Model



The VALIDATE partition has a Cumulative Lift of 5.16 in the 10% quantile (depth of 10) meaning there are 5.16 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TRAIN partition has a Cumulative Lift of 5.16 in the 10% quantile (depth of 10) meaning there are 5.16 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TEST partition has a Cumulative Lift of 5.04 in the 10% quantile (depth of 10) meaning there are 5.04 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

Cumulative lift is calculated by sorting each partition in descending order by the predicted probability of the target event P_ClassSIRA, which represents the predicted probability of the event "SIRA" for the target Class. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. The cumulative lift for a particular quantile is the ratio of the number of events across all quantiles up to and including the current quantile to the number of events that would be there at random, or equivalently, the ratio of the cumulative response percentage to the baseline response percentage. The cumulative lift at depth 10 includes the top 10% of the data, which is the first 2 quantiles, which would have 10% of the events at random. Thus, cumulative lift measures how much more likely it is to observe an event in the quantiles than by selecting observations at random.